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MASTER'S THESIS

Invasion and impact of *Percnon gibbesi*: Modeling herbivory-driven halo formation in Mediterranean rocky ecosystems

Sylvain Sylvestre Icaro Hamers

Master's Degree in Physics of Complex Systems

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Thesis Supervisor's Name: Tomás Sintés Olives

Thesis Supervisor's Name: Eva Llabrés Llambías

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Abstract

This thesis studies the environmental impact of the invasive crab *Percnon gibbesi* on Mediterranean sea ecosystems. To quantify this impact, a spatial seaweed-herbivore model coupling vegetation growth dynamics with an agent-based framework is developed and calibrated to field data. The ecosystem's stability is evaluated through the analysis of three variables: the ecosystem's sensitivity to macroalgae net primary productivity and diffusion rate; the characterization of halo size produced by the grazing of individual crabs; and the ecosystem's resilience to projected mean temperature increases. Simulations reveal a high sensitivity to macroalgae net primary productivity. Halos are shown to be a result of group dynamics rather than individual grazing. Alarming signs are raised as *Percnon gibbesi*'s grazing is shown to weaken the ecosystem's resilience to global warming, driving both flora and fauna to extinction within the coming decades.

Resumen

Esta tesis estudia el impacto ambiental del cangrejo invasor *Percnon gibbesi* en los ecosistemas del mar Mediterráneo. Para cuantificar este impacto, se desarrolla un modelo espacial alga-herbívoro que acopla las dinámicas de crecimiento de la vegetación con un modelo basado en agentes, calibrado con datos de campo. La estabilidad del ecosistema se evalúa mediante el estudio del comportamiento de tres variables: la sensibilidad del ecosistema a la productividad primaria neta y la tasa de difusión de las macroalgas; la caracterización de los halos producidos por el pastoreo individual de los cangrejos y la resiliencia del ecosistema ante los aumentos proyectados de la temperatura media de la superficie del mar. Las simulaciones revelan una alta sensibilidad a la productividad primaria neta de las macroalgas. Se observa que los halos son el resultado de dinámicas de grupo más que del pastoreo individual. Surgen señales de alarma al evidenciarse que el pastoreo de *Percnon gibbesi* debilita la resiliencia del ecosistema frente al calentamiento global, conduciendo tanto a la flora como a la fauna a la extinción en las próximas décadas.

Resum

Aquesta tesi estudia l'impacte ambiental del cranc invasor *Percnon gibbesi* en els ecosistemes marins del Mediterrani. Per quantificar aquest impacte, s'ha desenvolupat i calibrat un model espacial d'algues i herbívors que acobla la dinàmica de creixement de la vegetació amb un model basat en agents pels herbívors. L'estabilitat de l'ecosistema s'avalua mitjançant l'estudi de tres variables del model: la sensibilitat de l'ecosistema a la productivitat primària neta de les macroalgues i a la velocitat de difusió; la caracterització dels halos de generats per la pastura dels crancs i la resiliència de l'ecosistema a l'augment projectat de la temperatura mitjana de la superfície del mar. Les simulacions revelen una alta sensibilitat a la productivitat primària neta de les macroalgues. Es demostra que els halos són el resultat de la dinàmica de grup més que de l'alimentació individual. S'alcen senyals d'alarma, ja que es demostra que l'alimentació de *Percnon gibbesi* debilita la resiliència de l'ecosistema al canvi climàtic, conduint tant la flora com la fauna a l'extinció en les pròximes dècades.

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1 Introduction

Recent factors, such as increasing aquatic temperatures and human-mediated transport including shipping and ballast water, have allowed the introduction of well-suited herbivores into foreign ecosystems (Katsanevakis et al. (2014)). These invasive species pose a growing threat to native aquatic ecosystems worldwide as the typical lack of natural predators lead to mass grazings of the local flora (Pizarro-Borrull et al. (2025)). Mediterranean sea ecosystems are especially vulnerable to these invasions as recent studies report approximately 1000 exotic marine species settled in the waters (Galanidi et al. (2023)). The subtropical crab *Percnon gibbesi* is found among these invaders, first sighted in the Mediterranean sea in 1999 (Relini et al. (2000) ; Garcia (2000)), this species has spread rapidly across the rocky ecosystems native to these waters and have become a mayor threat to local ecosystem balance (Pizarro-Borrull et al. (2025)).

Percnon gibbesi, commonly called the **urchin crab** (figure 1), is a species of crab commonly found on both sides of the Atlantic Ocean and on the Pacific coast of North America (Manning and Holthuis (1981)). It's diet mainly consists on macroalgae, the most consumed species being *Haliptilon virgatum*, *Caulerpa cylindracea*, *Halopteris scoparia*, *Padina pavonica*, *Ulva compressa* and *Halimeda incrassata* (Pizarro-Borrull et al. (2025)). These crabs have a per capita grazing rate of 3.83 ± 1.71 WW g crab⁻¹ day⁻¹, ingesting from 0.1% to 10.9% of the total macroalgae production of the area that they inhabit (Pizarro-Borrull et al. (2025)). The **urchin crab** likes to shelter in rocky areas, grazing food around their home, this behaviour has been observed to contribute to the formation of reef halos, distinct circular zones of cleared vegetation surrounding patch reefs or rocky shelters, formed by the localized grazing of herbivores seeking refuge from predators (Figure 2).

Despite the threat that this invasive species have presented to the ecosystems of the Mediterranean for 20 years, the studies on their diets, grazing rates and population densities are only recent. This new information allows the development of ecosystem simulations that can help measure the scope of the impact that the invasive species has on the local flora. Seagrass-herbivore interaction models have already been successfully implemented to capture the spatial complexity of reef halos (Llabrés et al. (2025)), giving insight into how reef systems respond to environmental pressures.



Figure 1: Photograph taken of *Percnon gibbesi* hiding underneath a sea urchin. Taken by Danita Delimont (Delimont (2026)).

Crabs are one side of the story, the other side belongs to the seaweed that they feed on. There are multiple studies that have simulated vegetation growth through numerical models (Duarte and Ferreira (1997) ; Ruiz-Reynés et al. (2020) ; Sintes et al. (2005) ; Ruiz-Reynés et al. (2017)), successfully reproducing seaweed emerging patterns and dynamics. Even if there has been little research done on the specific species included in *Percnon gibbesi*'s diet, macroalgae has been thoroughly studied around the world providing huge datasets of seaweed Net Primary Productivity (NPP) and total biomass (Pessarrodona et al. (2022) ; Foster et al. (2025)) that are useful to create estimations on the growth rate of the affected algae species.

The aim of this project is to incorporate the data fetched by Pizarro-Borrull's team on *Percnon gibbesi* into a simulated ecosystem where the main actors are macroalgae and crabs. This research develops a spacially explicit numerical model that couples a simplified version of the equation proposed by Ruiz-Reynés et al. (2020) to model the growth of macroalgae with a crab agent-based framework akin to the one used in Llabrés et al. (2025). The project is divided in the following objectives:

- **Developing a general seagrass-herbivore model** with the aforementioned methods.
- **Tuning the model's parameters** with the information provided by *Percnon gibbesi* experts and macroalgae datasets, making sure that it correctly mirrors the behaviours currently being observed in nature.
- **Analysing the ecosystem's sensitivity** to different algae growth parameters, as the values retrieved from the dataset are an estimation of the real field behaviours.
- **Characterizing halo size** produced by *Percnon gibbesi* when modulating grazing rates and distances.
- **Evaluating the ecosystem's resilience to climate change** under different scenarios provided by The Intergovernmental Panel on Climate Change (IPCC (2023)) by introducing a mean temperature increase factor and a temperature dependence on the macroalgae mortality rate.



(a)



(b)

Figure 2: Left: Coral reef patches surrounded by halos of sand produced by herbivore members of a patch assemblage (fish, mollusks, and sea urchins) who remove sea grass and algae. Taken by CNES/Astrium (X-Ray Mag (2019)). Right: halo produced around a sea urchin. Image taken from: Pessarrodona et al. The Bulletin of the Ecological Society of America 100(3):e01544 (2019).

These objectives give insight into the dynamics between *Percnon gibbesi* and macroalgae in the context of the Mediterranean and provide a clearer picture on the impact that these herbivores have on native ecosystems.

2 Methods

All computer simulations were written in Julia language in the Google Colab environment. IFISC's supercomputer Nuredunna was utilized for the execution of programs. All code written can be found in the appendix.

2.1 Modeling macroalgae

2.1.1 Growth model

A simplified version of the model found in [Ruiz-Reynés et al. (2020)] is used to simulate the growth and spread of the algae. We derive the model starting from the basic growth equation:

$$n_{i,j}(t + \delta t) = n_{i,j}(t) + \delta t n_{i,j}(t)(\omega_b - \omega_d), \quad (1)$$

with $n_{i,j}(t)$ the biomass at a given lattice site $\vec{x} = (i, j)$ [gr DW (Dry Weight)]; ω_b and ω_d the algae birth and mortality rates, respectively [day^{-1}].

We assume a local interaction that is introduced through a density dependent mortality rate that is enhanced in low and high density regions:

$$\omega_d = \omega_{d0} + a\hat{n}_{i,j}(t)(\hat{n}_{i,j}(t) - 1) \quad (2)$$

with $\hat{n}_{i,j}(t) = n_{i,j}(t)/(\Delta L)^2 \rho_{max}$, ρ_{max} the algae density carrying capacity; $(\Delta L)^2$ the area of a lattice in our simulation grid, and ω_{d0} the intrinsic mortality rate.

Therefore local biomass density evolves in time following the equation:

$$\frac{dn_{i,j}(t)}{dt} = n_{i,j}(t) (\omega_b - \omega_{d0} + a\hat{n}_{i,j}(t)(1 - \hat{n}_{i,j}(t))) \quad (3)$$

That can be written in reduced units as:

$$\frac{d\hat{n}_{i,j}(t)}{dt} = \omega_0 \hat{n}_{i,j}(t) + a\hat{n}_{i,j}^2(t) - b\hat{n}_{i,j}^3(t) \quad (4)$$

with $\omega_0 = \omega_b - \omega_{d0}$.

Each term captures a specific, density-dependent driver of macroalgal population dynamics:

- $\omega_0 \hat{n}_{i,j}(t)$ is the intrinsic baseline growth term.
- $a\hat{n}_{i,j}^2(t)$ is the cooperative growth term, which mathematically introduces an Allee effect.
- $b\hat{n}_{i,j}^3(t)$ is the competitive mortality term that enforces local spatial saturation.

To make the model more general, different coefficients have been assigned to the cooperation and competitive terms in eq. 4.

Since eq. 4 is a local equation, to allow the spreading of the algae, we include a diffusive term in eq. 4 similarly as it is done in the Fisher–Kolmogorov–Petrovsky–Piskunov model (Murray (2002)).

$$\frac{d\hat{n}_{i,j}(t)}{dt} = \omega_0 \hat{n}_{i,j}(t) + a\hat{n}_{i,j}^2(t) - b\hat{n}_{i,j}^3(t) + \epsilon \hat{\nabla}^2 \hat{n}_{i,j}(t) \quad (5)$$

Note: $\nabla^2 n = (\Delta L)^2 \rho_{max} \nabla^2 \hat{n} = \rho_{max} \hat{\nabla}^2 \hat{n}$, with $\hat{\nabla}^2 = (\Delta L)^2 \nabla^2$. ρ_{max} included in ϵ .

2.1.2 Grazing

Inspired by [Llabrés et al. (2025)], the impact of *Percnon gibbesi* on algae is represented with an additional grazing term in the algae growth model. Within the study area crabs shelters are located at positions $\vec{x}_k$, and we assume they are able to move away from the shelter an average distance σ following a 2-d gaussian distribution:

$$\mathcal{F}(\vec{x}, t) = \frac{\alpha}{2\pi\sigma^2} \sum_{k=1}^{n_c(t)} e^{(\vec{x}-\vec{x}_k)^2/2\sigma^2} \quad (6)$$

with α the individual grazing rate [gr DW day⁻¹ crab⁻¹]. We also assume that each shelter can hold a single crab maximum, that is, $n_c(\vec{x}_k, t) = \{0, 1\}$.

In a time step δt [day] an amount of biomass $\delta t \mathcal{F}(\vec{x}, t) (\Delta L)^2$ [gr DW] is subtracted from the lattice site located at $\vec{x} = (i, j)$ of area $(\Delta L)^2$. Introducing the grazing effect, normalized by the algae carrying capacity ρ_{max} , into eq. 5, the equation for the algae growth is given by:

$$\frac{d\hat{n}_{i,j}(t)}{dt} = \omega_0 \hat{n}_{i,j}(t) + a\hat{n}_{i,j}^2(t) - b\hat{n}_{i,j}^3(t) + \epsilon \hat{\nabla}^2 \hat{n}_{i,j}(t) - \frac{\mathcal{F}_{i,j}(t)}{\rho_{max}} \quad (7)$$

2.2 Modeling the crab population

2.2.1 Breeding

Mature female crabs have from 2 to 4 major reproductive events in a season. It has been estimated up to 99,9% loses because of egg hatching, fungi and bacterial infections, environmental stress, starvation and predation of larvae and juvenile individuals by fishes, octopus and larger crabs. Each mature female crab lays of the order of 10^4 eggs during a breeding event, thus, only 10 young individuals will finally settle down. Larvae are dispersed via currents a long distance, thus, within the simulated ecosystem they can settle in any randomly chosen empty lattice site.

The characteristic lifespan of *P. gibbesi* is about 3 years. They are able to reproduce once maturity is reached (after the first year). Thus, the number of new individuals can be estimated from:

- The total number of crabs at any given time t : $n_c(t)$
- Female fraction: 1/2
- Mature fraction: 2/3
- Average of major breed events in a season: 3

- Survival of juvenile crabs in a breeding event per mature female: 10

So, the number of new crabs in a season (per year) can be estimated as:

$$\gamma(t) = n_c(t) \times \frac{1}{2} \times \frac{2}{3} \times 3 \times 10 = n_c(t) \times 10 \text{ [crabs } yr^{-1}]$$

and, within a time δt [days]: $\gamma \delta t / 365$.

Reproduction can be linked to resources abundance and to the density carrying capacity of the crab population: $\rho_{c,max}$. The role of the carrying capacity is given by the factor:

$$\zeta(t) = \max \left(0, 1 - \frac{n_c(t)}{\rho_{c,max} L^2} \right) \quad (8)$$

with $n_c(t) = \sum_{k=1}^{n_c(t)} n_c(\vec{x}_k, t)$, the total crab population.

Let $\mathcal{G}(\vec{x}_k, t)$ be the total biomass available to a crab located at $\vec{x}_k$ within a distance σ , being the lattice sites visited according to a gaussian distribution:

$$\mathcal{G}(\vec{x}_k, t) = \frac{(\Delta L)^2}{2\pi\sigma^2} \sum_{i,j/|\vec{x}_{i,j}-\vec{x}_k| \leq \sigma} e^{(\vec{x}_{i,j}-\vec{x}_k)^2/2\sigma^2} \hat{n}_{i,j}(t) \quad (9)$$

The breeding probability of a crab located at $\vec{x}_k$ is:

$$\mathcal{P}(\vec{x}_k, t) = \mathcal{G}(\vec{x}_k, t) \zeta(t) \quad (10)$$

A crab (mature female) located at $\vec{x}_k$ breeds if $\eta < \mathcal{P}(\vec{x}_k, t)$, with $\eta \in [0, 1)$ a flat random number. In this case, up to $10 \times \delta t / 365$ new individuals could be added to the system for each successful breed. A summation is performed over all crabs: $0 \leq k \leq n_c(t)$.

2.2.2 Mortality

Let μ [years] be the characteristic lifespan of the crabs. The same way that breeding is linked to resource abundance $\mathcal{G}(\vec{x}_k, t)$, mortality also is. Therefore mortality is computed for each individual crab in the study area following the expression:

$$\mu_k(\vec{x}_k, t) = \mu \cdot \max \left(0, 1 - e^{-\mathcal{G}(\vec{x}_k, t)} \right) \quad (11)$$

The mortality probability of a crab located at $\vec{x}_k$ is:

$$\mathcal{M}(\vec{x}_k, t) = \frac{\delta t}{365 \cdot \mu_k(\vec{x}_k, t)} \quad (12)$$

In each time step δt a crab located at $\vec{x}_k$ dies if $\eta < \mathcal{M}(\vec{x}_k, t)$, with $\eta \in [0, 1)$ a flat random number.

2.3 Parameter choice

The simulations are dependant on the parameters that we choose, so the selection of these parameters is a crucial step that needs proper justification. The type of parameters can be separated into different categories: General simulation parameters, Algae parameters and *Percnon gibbesi* parameters.

2.3.1 General simulation parameters

The value of these parameters aren't bound by measurements and can be chosen freely as they don't necessarily influence the results. It is however important to choose parameters that allows us to observe the impact of the crabs in the fauna while not making the simulations unnecessarily heavy to compute.

The area of study has been set to $50 \times 50 \text{ m}^2$ in order to observe the large-scale impact of the crabs. The grid spacing has been set to 0.1 m meaning that every lattice unit is 1 dm^2 which makes the grid smoother. The last general parameter is the time step which has been set to $\delta t = 0.1$ day allowing the model to evolve smoothly.

2.3.2 Algae Parameters

Tuning the parameters of the algae has proven to be the most challenging section of the project due to the scarcity of data available.

A reasonable value for the algae density carrying capacity is estimated to $\rho_{max} = 100 \text{ [g DW m}^{-2}\text{]}$ (Ruitton et al. (2005)), which leads to 1 g of dry algae per lattice cell. The conversion rate from wet to dry algae weight is generally accepted to be: $WW = DW \times 10$. Both are reasonable general estimations based on observation.

The algae model follows eq. 5 and is bounded by the values of the parameters ω_0 , a , b and ϵ . The first three values determine the dynamics of the growth in individual lattice cells accounting for local interactions alone. The choice of the parameters has been done to ensure the plant growth is located in a bistable regime that is characterized by a rich dynamics. The selected values are:

- $\omega_0 = -0.4$
- $a = 1.4$
- $b = 1$

Thus, there are three fixed points in the solution of Eq. 4, that are illustrated in figure 3:

- $\hat{n}_{i,j}(t) = 0$ Stable fixed point
- $\hat{n}_{i,j}(t) = 0.4$ Unstable fixed point
- $\hat{n}_{i,j}(t) = 1$ Stable fixed point

As a consequence, the model behavior is determined by the initial biomass:

- Initial biomass under 0.4 will slowly drive the algae to death.
- Initial biomass between 0.4 and 1 will make the algae grow until reaching a biomass of 1.

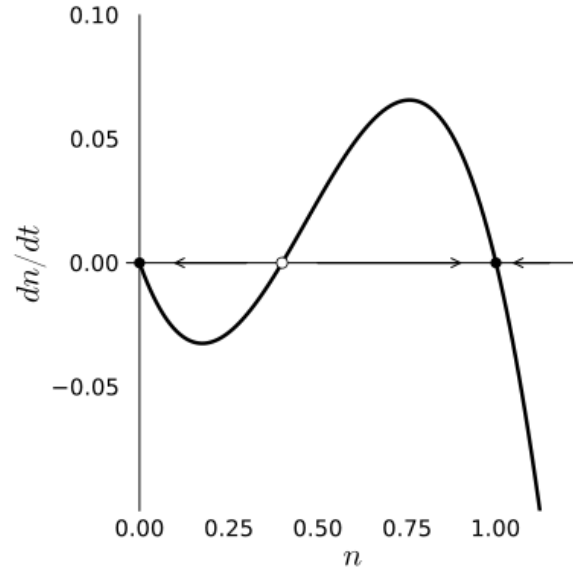


Figure 3: Phase portrait of the algae growth model.

- Initial biomass over 1 means the algae is saturated and therefore reduces to the stable carrying capacity of 1.

Without affecting the dynamics, we can multiply the model by a scalar to tune the speed at which the algae grows and decays. In order to determine the scaling factor, we looked into **A global dataset of seaweed net primary productivity** (Pessarrodona et al. (2022)). Here we find the NPP (Net Primary Productivity) of genus *Padina* and *Caulerpa* which correspond to the 2nd and 4th most consumed seaweed by the *Percnon gibbesi*. The original data measures the productivity of these seaweed species in kilograms of carbon per squared meter and year. Rescaling these values into average NPP in grams of dry weight per dm^2 and per day, using the available conversion rate from carbon to dry weight, and computing the average over all *Padina* and *Caulerpa* variations we get an average Net Productivity Rate of $0.0023 \text{ g DW dm}^{-2} \text{ day}^{-1}$.

As seen on figure 3, dn/dt corresponds to the speed of growth of the algae. The average growth rate of the model can be estimated with $[\text{Area} = \text{Average growth} \times \text{width}]$. Only the zone between the fixed points 0.4 and 1 will be taken into consideration for the measurement since it corresponds to the growing zone of the model. The scaling factor can be derived from:

$$\int_{0.4}^1 (\omega_0 \hat{n} + a\hat{n}^2 - b\hat{n}^3) d\hat{n} = 0.0252 \quad (13)$$

$$\text{Average growth} = \frac{0.0252}{0.6} = 0.042 \quad (14)$$

we get an average growth of $0.042 \text{ g DW dm}^{-2} \text{ day}^{-1}$.

In order to reach the observed NPP of $0.0023 \text{ g DW dm}^{-2} \text{ day}^{-1}$ without affecting the dynamics of the system we just need to rescale the algae growth model by a constant factor. In our case this factor is: $c = 0.0023/0.042 = 0.055$.

The last parameter to consider is the spatial diffusion coefficient (ϵ). Due to the absence of field data, a parameter estimation was conducted via a systematic parameter sweep across the range $\epsilon \in [0.01, 0.20]$. Simulations were executed on a $100 \times 100 \text{ dm}^2$ domain initialized with a central $40 \times 40 \text{ dm}^2$ algae patch. To assess the spatial propagation, the daily specific biomass accumulation along the expanding frontier was tracked and averaged over a one-year simulation period. The optimal parameter was identified as $\epsilon = 0.07$, which achieved the closest agreement with the target empirical mean growth rate of $0.0023 \text{ g DW dm}^{-2} \text{ day}^{-1}$.

The final algae growth model is:

$$\frac{d\hat{n}_{i,j}(t)}{dt} = 0.055 \cdot [-0.4 \cdot \hat{n}_{i,j}(t) + 1.4 \cdot \hat{n}_{i,j}^2(t) - \hat{n}_{i,j}^3(t)] + 0.07 \cdot \nabla^2 \hat{n}_{i,j}(t) \quad (15)$$

2.3.3 *Percnon gibbesi* parameters

The behavior of *Percnon gibbesi* is largely studied in the article **Prominent grazing rates and feeding preferences of an abundant exotic benthic herbivore in the Mediterranean Sea** (Pizarro-Borrull et al. (2025)) allowing us to set precise values to the crab parameters that control their growth.

The mean *Percnon gibbesi* density carrying capacity is $\rho_{c,max} = 100 \pm 72$ [crabs in 100 m^2] this, coupled with the study size of $50 \times 50 \text{ m}^2$ tells us that there can be a maximum of 2500 crabs. The average *Percnon gibbesi* grazing rate is $\alpha = 3.83 \pm 1.71$ [g WW crab $^{-1}$ day $^{-1}$]. Even though the characteristic grazing distance has not been directly measured, Pizarro-Borrull et al. (2025) have confirmed a maximum value of $\sigma = 1 \text{ m}$.

2.4 Sensitivity analysis on the parameter choice

Given the high uncertainty in empirical field measurements, we focused the study on sensitivity analyses to evaluate the impact of parameter selection on the model's dynamics. This includes:

- The effect of the net primary productivity (NPP) on the population dynamics.
- The effect on the size of the grazing halos formed around *Percnon gibbesi* shelters as a function of the crab grazing rate, α , and grazing distance, σ .
- The impact on the algae mortality rate due to the ocean warming.

2.4.1 Influence of NPP and spatial diffusion on population dynamics

The main parameters that can be manipulated without removing the bi-stability of the model are the Net Primary Productivity (NPP) which is regulated by the scaling factor, c , and the spatial diffusion coefficient, ϵ .

We have found that the model is very sensitive to the NPP since it determines whether the algae can grow faster/slower than the amount of biomass removed due to crab grazing. We have varied the scaling factor, c , up to $\pm 40\%$ of its averaged observed data. The range of values selected are $c \in [0.033, 0.044, 0.055, 0.066, 0.077]$ corresponding to $\text{NPP} \in [0.14, 0.18, 0.23, 0.28, 0.32]$. The model is

found to be less sensitive to the diffusion parameter that has varied two orders of magnitude in the interval: $\epsilon \in [0.007, 0.014, 0.07, 0.35, 0.7]$.

The model is simulated for a time period of 30 years where the algae and *Percnon gibbesi* densities are recorded for each time step. Each simulation starts with an initial condition of 80 ± 10 [g DW m⁻²] macroalgae density and 20 crabs to avoid early extinctions. The model presents randomness in the shape of the crabs dynamics (reproduction, death and placement), therefore 10 simulations has been run for each combination of NPP and ϵ . The values measured are:

- Final algae biomass density (g DW m⁻²)
- Standard deviation of the algae biomass density
- Final *Percnon gibbesi* density (in 100 m²)
- Standard deviation of the *Percnon gibbesi* density
- Final biomass to crab ratio (g DW m⁻² crab⁻¹)

The simulations for the pair of parameters [NPP = 18, $\epsilon = 0.7$], [NPP = 23, $\epsilon = 0.007$], [NPP = 23, $\epsilon = 0.014$] and [NPP = 23, $\epsilon = 0.07$] were extended to 100 years to reach ecosystem stability.

Graphs showcasing the average trajectory of the algae and crab densities for each combination of parameters are also given in the Appendix section.

2.4.2 Size of the grazing Halo

Grazing is modelled to make crabs feed of algae in a continuous manner around them following a Gaussian distribution. When the crab grazing is higher than the speed at which the algae can be restored, a round halo of empty space forms around them. The size of these halos are determined by the average *Percnon gibbesi* consumption, α , and the characteristic grazing distance, σ .

The grazing halo size is computed for values of σ ranging from 10 cm to 50 cm. We have also computed values for the grazing rate, α , ranging from 0.41 g WW day⁻¹ to 7.25 g WW day⁻¹, twice the standard deviation (Pizarro-Borrull et al. (2025)) while maintaining a σ of 30 cm to visualize the halos.

The simulations where run for a time period of 1 year in a 2×2 m² grid with spacing 0.01 m, this means that each lattice unit is 1 cm² which increases the precision of the halo size computation.

2.4.3 Global warming effect on long-term projections

In all the cases studied, the model reaches an equilibrium characterized by the coexistence between *Percnon gibbesi* and algae. Unfortunately, algae prosperity is heavily affected by temperature. To study the long term behaviour of the ecosystem we must introduce the effect of the temperature on the algae growth rate. This is done through a temperature dependent algae mortality rate, as it was previously introduced in (Llabrés et al. (2025)). Thus, the new intrinsic mortality rate, ω_{d0} , is replaced by a temperature dependent factor:

$$\omega_{d0} \rightarrow \omega_d(T) = \xi(T - T_0) + \omega_{di} \quad (16)$$

where T is the surface sea water temperature, $T_0 = 25^\circ C$. Due to the limited field data, we have chosen $\xi = 0.0012 \text{ day}^{-1}^\circ C^{-1}$ as the one found for seagrasses in Kāneʻohe Bay, Oʻahu, Hawaiʻi, taken from Innes-Gold et al. (2025). ω_{di} is the initial intrinsic mortality rate implemented in the model.

Since the temperature is assumed to increase linearly with time, with a growth rate that depends of the global warming scenario, $\Delta T = T - T_0 = \beta t$, we can rewrite the change of the algae mortality rate as function of time:

$$\omega_d(t) = \omega_{di} + \Upsilon t \quad (17)$$

with $\Upsilon = \xi\beta$.

The Intergovernmental Panel on Climate Change (IPCC) provides multiple predictions of temperature increase over the century (IPCC (2023))¹. Three main scenarios with different levels of human emissions are presented by this panel; they forecast the following temperature increases by the end of the century, from which β is obtained:

- SSP1-1.9: $1.4^\circ C \rightarrow \beta = 6.2 \cdot 10^{-5} \text{ }^\circ C \text{ day}^{-1}$
- SSP2-4.5: $2.7^\circ C \rightarrow \beta = 1.2 \cdot 10^{-4} \text{ }^\circ C \text{ day}^{-1}$
- SSP5-8.5: $4.4^\circ C \rightarrow \beta = 1.9 \cdot 10^{-4} \text{ }^\circ C \text{ day}^{-1}$

Implementing an update system following equation 17, ten simulations spanning 150 years are computed for each one of the temperature forecast provided by the IPCC. A model simulating only the seagrass without crabs is also implemented to appreciate the impact that *Percnon gibbesi* has on the life span of the algae.

¹SSP stands for "Shared Socio-economic Pathway". The following number indicates the mitigation challenge, ranging from 1 (sustainable development) to 5 (fossil-fuel development). The final number is the radiative forcing in W/m^2 by year 2100, that can be translated into a temperature increase by the end of the century.

3 Results & discussion

3.1 Influence of NPP and spatial diffusion on population dynamics

The results of the sensitivity analysis on the algae Net Primary Productivity (NPP) and diffusion rate, ϵ , can be seen in figure 4. They show a high sensitivity on the NPP given that an increase of 40% over the original NPP value doubles the seaweed density (figure 4a) and almost triples the crab density (figure 4c). On the other hand, the model shows little sensitivity to the strength of the diffusion rate, ϵ , as an increase of two orders of magnitude produces just an increase of 12% of the algae density and a 24% increase of the crab density. The influence of ϵ becomes less relevant for higher values of NPP. The influence of ϵ is mostly noticeable in the case of $\text{NPP} = 0.23 \text{ [g WW crab}^{-1} \text{ day}^{-1}]$ where a big increase in both the algae and crab densities can be observed for intermediate values of ϵ .

The standard deviations of the final algae density (figure 4b) showcases high flora stochasticity in low algae productivity and high diffusion rate scenarios, peaking at 1.73 for $\text{NPP}=0.18$ and $\epsilon=0.7$; from these results we can conclude that flora stabilizes as algae productivity grows. While algae variability shows clear trends, *Percnon gibbesi*'s standard deviations don't (figure 4d). The time evolution graphs found in the appendix (Figure 14) hints us that, while algae variability is smooth over time, crab variability changes in a stochastic manner. It is an interesting phenomenon where the agent's population is volatile but it's effect on the system is predictable.

Both figures 4a and 4c showcase how high algae productivity benefits both species by increasing the biomass and the *Percnon gibbesi* population; however, figure 4e exposes that favourable conditions leaves each individual crab with less food to feed on, hitting a consistent minimum of 1.33 grams of seaweed per crab. This explains the stability seen at favourable parameters: a higher crab density leads to a higher consumption of resources, as a result, this process leads to less nutrients per crab that increases the crab mortality, balancing the ecosystem. The crab population is limited by available seaweed resources, leading to crab density being always lower than algae density. This behaviour can be clearly seen in the extra time series included in the appendix (figure 14). The high biomass to crab ratios found in un-favourable conditions can easily be explained by the low crab density leaving batches of ungrazed seaweed.

The snapshots taken of the ecosystem under the range of parameters studied (Appendix, Figure 9, 10) exposes a very prominent feature of the seagrass-herbivore model developed in this project. The randomness found within the newborn crab placements can lead the ecosystem into extreme outcomes. Under low macroalgae productivity scenarios, after approximately 10 years of *Percnon gibbesi* invasion, the ecosystem enters a critical low-density bottleneck where small population fluctuations can drive the ecosystem into three divergent trajectories:

- **Ecosystem collapse:** *Percnon gibbesi* manage to completely shave off all the local flora, leading to their consequential extinction by starvation.
- **Herbivore extinction and flora recovery:** Low food resources leads to crab extinction before they fully graze the algae, leading to it's slow recovery.
- **Dynamic Equilibrium:** The ecosystem populations balance each other leading to shifting halos.

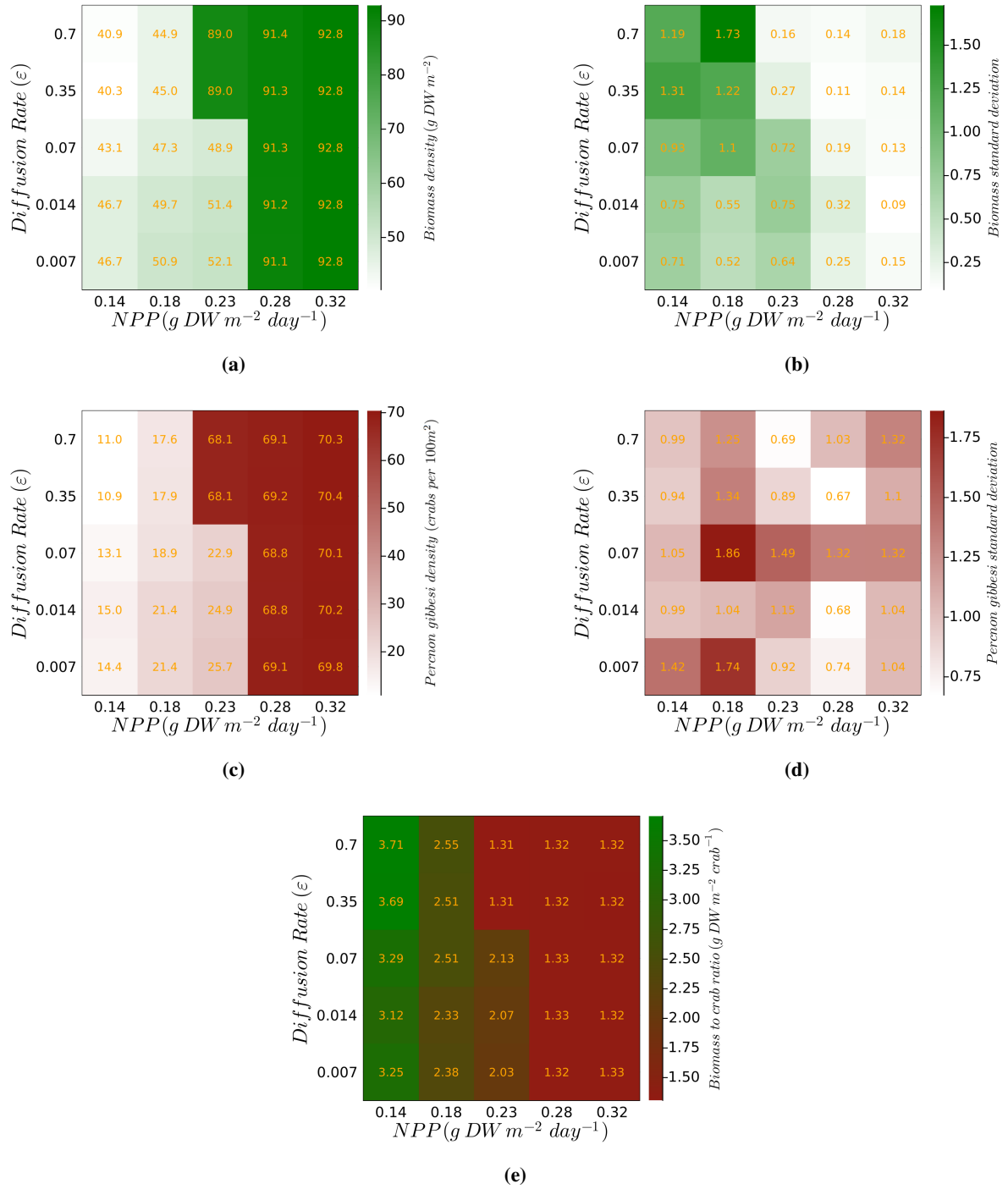


Figure 4: Heatmap graphs analysing the sensitivity of the ecosystem to different parameter values. The parameters analysed are the Net Primary Productivity (NPP) and the diffusion rate ϵ . The effects on the ecosystem analysed are the final algae biomass density (a), it's standard deviation (b); the final *Percnon gibbesi* density (c), it's standard deviation (d) and the algae to crab ratio (e).

3.2 Halo size

The size of grazing halos formed by *Percnon gibbesi* can be seen in figures 5 and 6. The halo's radius dependence on average consumption, α , and grazing range, σ , can be seen in figure 7.

The results show that a consumption rate threshold must be surpassed to create visual grazing halos, this value lies at around 2 g WW day⁻¹ (see Figure 7a). After surpassing this threshold, a logarithmic type of growth is observed with an early rapid increase of the halo size followed by a slow growth regime. This behaviour is expected considering that the available biomass roughly increases by a power of two with the grazing distance. The results also show that halo size can surpass the grazing range σ . Mathematically, this happens because a high grazing rate can push the lattice algae density to very small values that propagate to neighbour cells through diffusion. Biologically, this shows that the model naturally increases the grazing distance for hungry crabs, forming bigger halos than what the value of σ might suggest.

Changing the crab's grazing range σ also yields interesting results. The size of halos follow a reverse parabola, reaching a maximum size of 38 ± 1 cm radius within the range $21 < \sigma < 26$ cm (Figure 7b). The origin of this behaviour is found within the Gaussian-distributed grazing nature:

- A small σ intensifies the grazing close to the crab, allowing the grazing to surpass the threshold needed to form halos (Figure 6a).
- Increasing σ spreads the grazing over a larger distance, generating larger grazing halos (Figure 6b).
- The above behavior happens until σ surpasses 38 cm where the maximum grazing rate falls under the necessary threshold to create halos (Figure 6e).

These results showcases how *Percnon gibbesi*'s impact on the flora varies depending on the individual's eating habits. A high grazing rate α paired with a low grazing distance σ proves to be a specially lethal combination. However, this case is unlikely considering that expert's observations pointed to an average consumption rate of $\alpha = 3.83$ [g WW crab⁻¹ day⁻¹] and a mean grazing distance of 1 m, setting the halo size for the average crab diet to 0 cm.

Even if an individual crab is unlikely to create halos of empty seaweed, on a large scale simulations patches of empty space can be found spread through the study zone as seen in the snapshots taken of the sensitivity analysis (Appendix, Figures 9, 10, 11, 12, 13). This phenomenon happens when crabs gather close enough to each other and feed of the same patch of seaweed, incrementing the effective grazing rate of this particular zone.

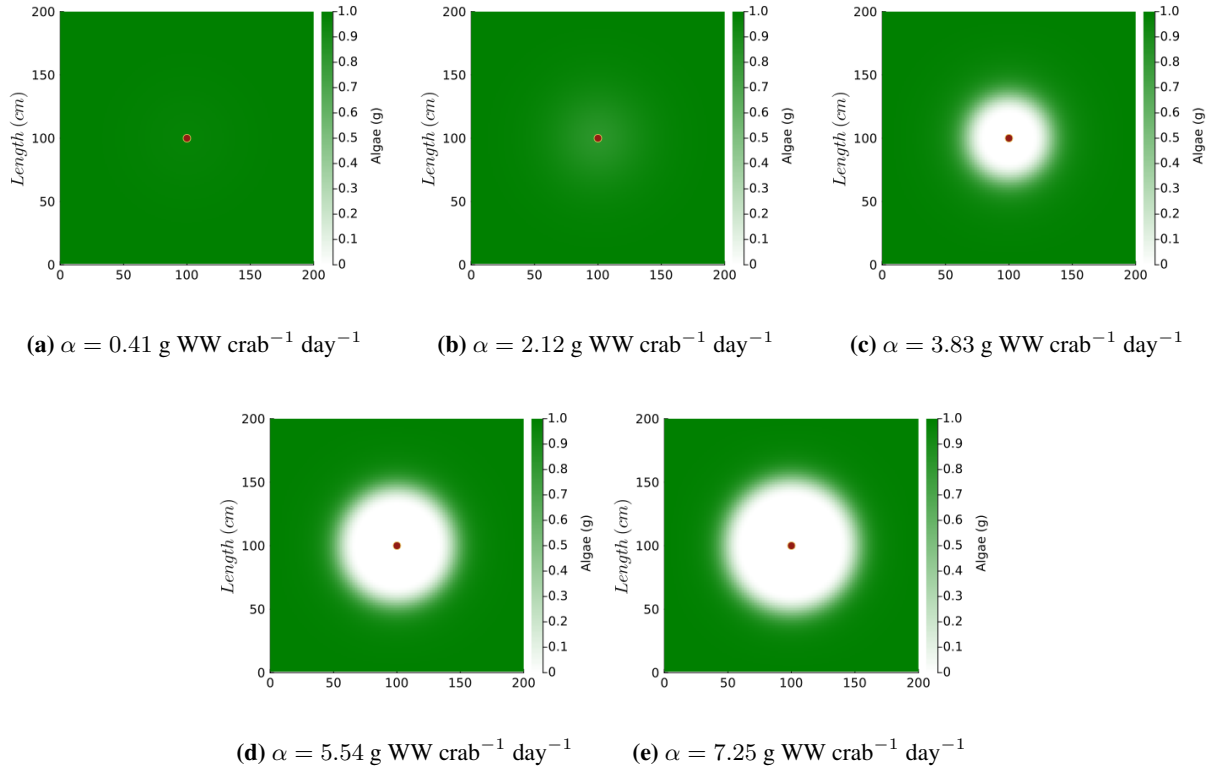


Figure 5: Snapshots of the model after 1 year of *Percnon gibbesi* grazing, varying the crab's average algae consumption α . The radius of these halos are: (a) 0 cm, (b) 0 cm, (c) 33 cm, (d) 46 cm, (e) 52 cm.

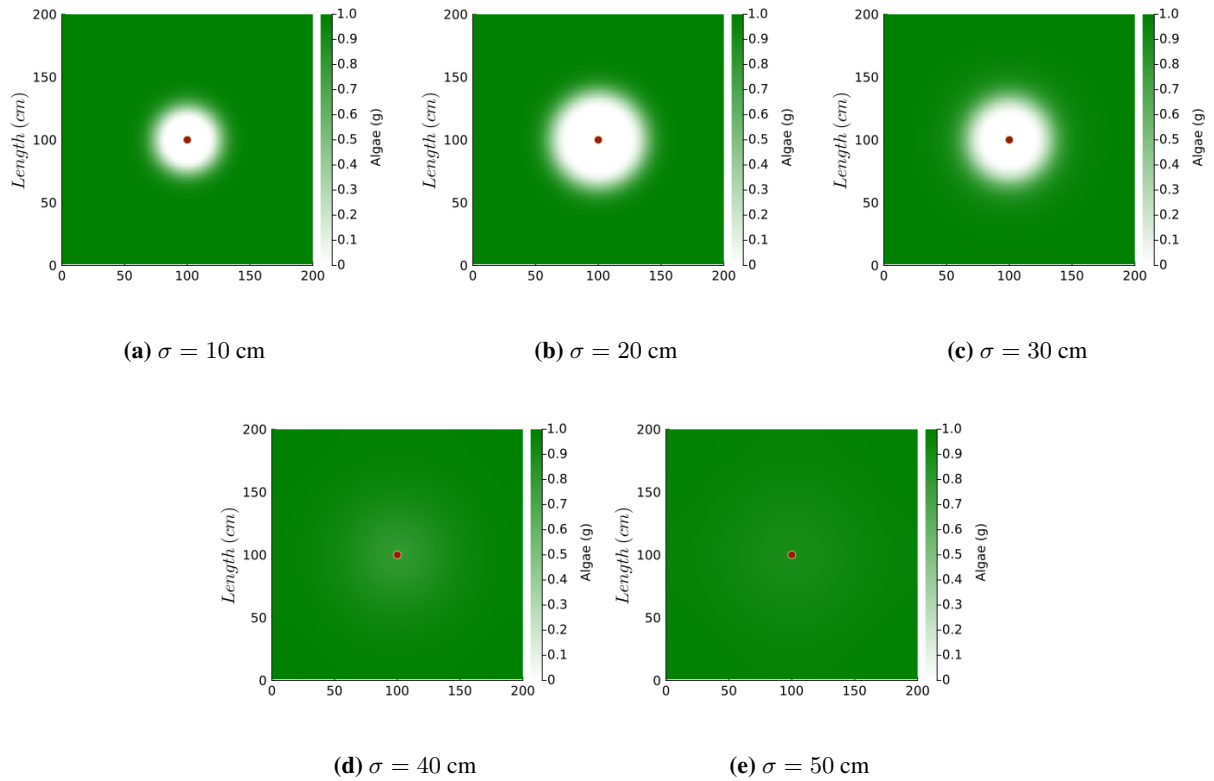


Figure 6: Snapshots of the model after 1 year of *Percnon gibbesi* grazing, varying the crab's grazing distance σ . The radius of these halos are: (a) 26 cm, (b) 37 cm, (c) 33 cm, (d) 0 cm, (e) 0 cm.

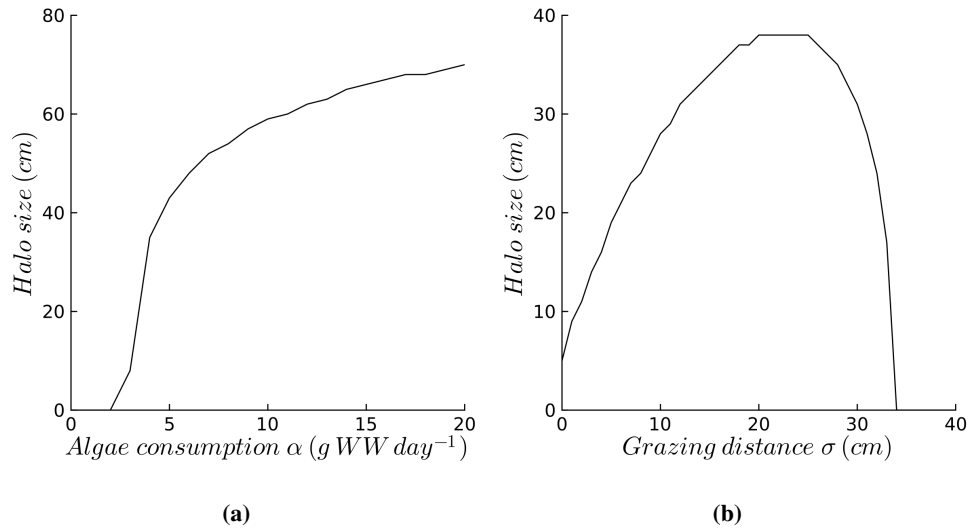


Figure 7: Graphs showcasing the relation between halo radius, average consumption (a) and grazing distance (b).

3.3 Global warming effect on long-term projections

The long-term projections of the seawater warming can be seen in figure 8. They expose how global warming will inevitably drive both the flora and fauna to extinction. The model predicts, at best case scenario (SSP1-1.9), a *Percnon gibbesi* extinction in 80 years and an algae extinction in 120 years. The life expectancy decreases with higher carbon emissions forecasts, crabs and algae dying in 45 and 70 years, respectively in the SSP2-4.5 scenario. In worst case scenario (SSP5-8.5), crabs go extinct after 30 years (in 2053 after the 2023 IPCC report) and algae after 47 years (in 2070). In all cases *Percnon gibbesi* die before algae showing how crabs cannot outlive their food source. All figures manifest a similar shape that contracts in time with the increase in carbon emissions; this shape can be divided in three distinct phases:

1. **Crab invasion:** In this phase a small group of crabs invade the study zone, the abundance of resources makes the population of crabs explode in size. Meanwhile the algae which starts at 80% capacity can grow up to the carrying capacity limit.
2. **Slow consumption:** This is the point in which crabs reach the saturation point driven by the food abundance. In this phase crabs slowly consume algae until they reach a balance between algae and *Percnon gibbesi* (around year 40 in SSP1-1.9).
3. **Slow decay:** This phase would present equilibrium in a world with zero emissions. However, increasing temperatures slowly decreases the algae net productivity, reducing the biomass recovery rate and driving crabs to extinction due to lack of resources. Even after the extinction of *Percnon gibbesi*, the flora cannot recover to its carrying capacity and instead slowly disappears from the sea.

These graphs also showcases the impact of *Percnon gibbesi* on algae as, without a herbivore, seaweed would survive 35% more on average, dying rapidly once the net productivity falls under the growth threshold. The damage caused by *Percnon gibbesi* is hard to recover as a cause of crabs eating new algae sprouts limiting its expansion possibilities.

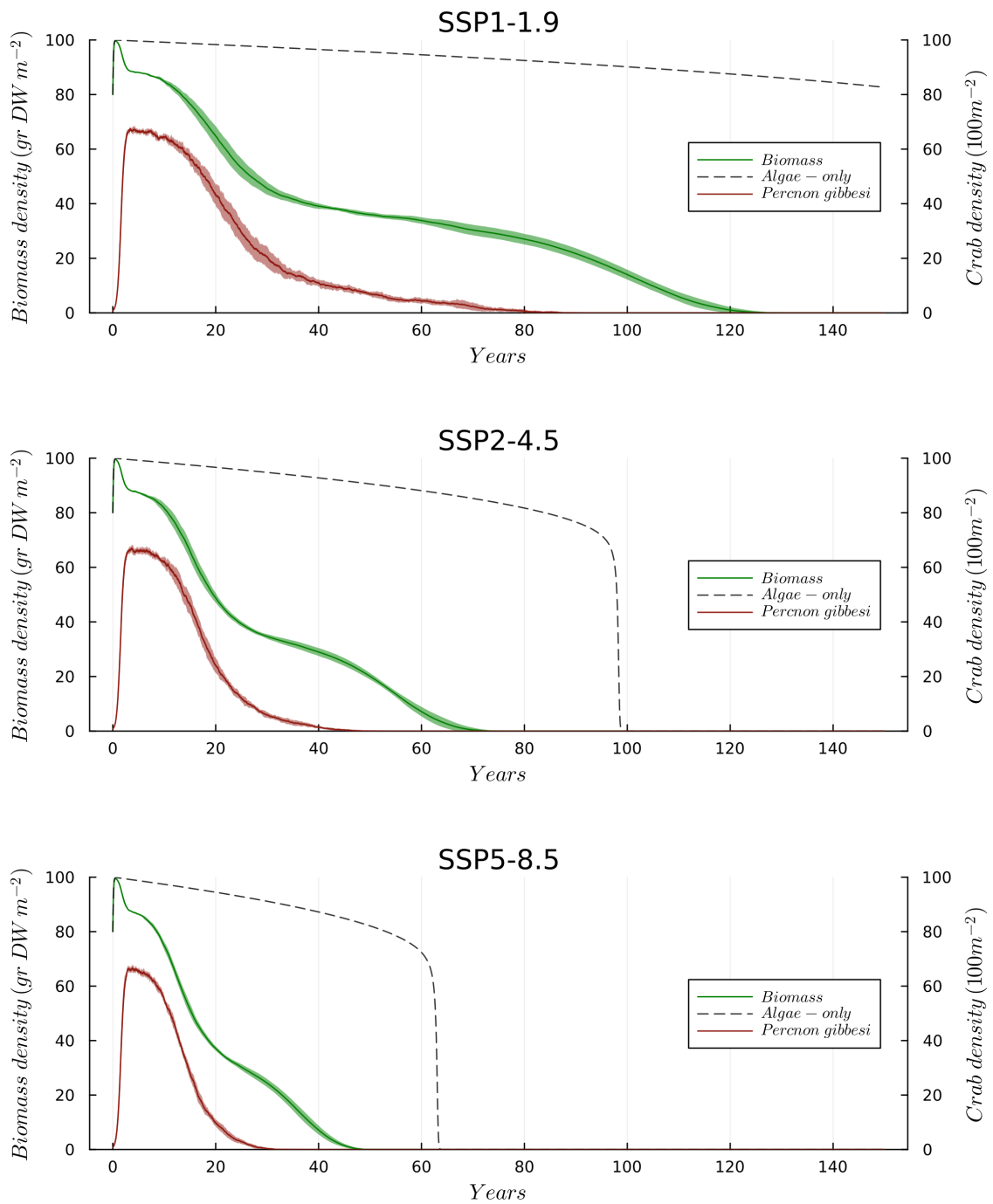


Figure 8: Temporal graphs showcasing the evolution of *Percnon gibbesi* density and algae density over the span of 150 years. Averaged over 10 runs with Standard deviations represented as ribbons around the mean value.

4 Conclusions

In this project we have successfully implemented a general seagrass-herbivore model that combines differential equation-based modelling with agent-based architectures. This approach has proven to be useful in simulating herbivore-resource dynamics in an intuitive and visually engaging manner. This master thesis has focused on its application to a specific Mediterranean ecosystem but the model can be easily generalized to different contexts by modifying the flora and fauna parameters.

The macroalgae growth equation has proven to be a challenging feature to adjust without macroalgae-specific data and presents the biggest room for improvement in future iterations of the *Percnon gibbesi*-seaweed model. In order to reduce this shortcoming, a sensitivity analysis has been performed to analyse the ecosystem behavior under a broad range of outcomes. This analysis has exposed the direct impact of the Net Primary Productivity of the algae onto the ecosystem dynamics, being the main component to drive the system into a high or low density equilibrium values.

The halo formation and the characterization of its size was successfully analysed in outlier *Percnon gibbesi* conditions, appearing more frequently as a result of group dynamics. The analysis highlights the size dependency on grazing rates and distances, exposing the algae bistability dynamics underlying the phenomena.

We have finally investigated the impact of the global warming on the population dynamics of both, the algae and the *Percnon gibbesi* species. It has been assumed a linear relationship between the algae mortality rate and the temperature, that also grows linearly with time. The results under the different scenarios present in the IPCC-2023 report, from the mildest case, SSP1-1.9, to the worst one, SSP5-8.5, allow us to take a glimpse into the unfortunate future of the *Percnon gibbesi* fauna. Any steady increase in temperature leads both crab and macroalgae to extinction. The results also showcase the damage done by *Percnon gibbesi* as the estimated lifetime of the affected flora reduces significantly and struggles to recover even long after the crabs become extinct.

It is important to keep in mind that, as the seagrass-herbivore model developed presents randomized elements, the results analysed in this project are mean estimates averaged across many possible outcomes. This is to say that the results presented in these papers should not be taken as grounded facts, unlucky circumstances can quickly drive the ecosystem into extinction much quicker than expected.

Further improvements to the model could be made by implementing seasonal changes in temperature, leading to seasonally fluctuating population dynamics. Crab's grazing rates and distances could also be randomized following a Gaussian distribution derived from field data variance. We believe that these changes would depict the ecosystem more accurately and are worth exploring in future works.

Overall, the results obtained provide a wider view on the multifaceted impacts that the invasive *Percnon gibbesi* have on the Mediterranean ecosystem. Our findings, properly delivered to marine scientists, can support targeted conservation measures to sustain Mediterranean's rich and diverse life.

Appendix

Code availability

All the code can be found in the following GitHub repository:

https://github.com/SylvainHam/TFM-Invasion_and_impact_of_Percnon_gibbesi/tree/main

AI disclaimer

Google's **Gemini** and Anthropic's **Claude** tools have been used to debug code, improve writing and translate the abstract into Catalan.

Extra figures

Ecosystem development

Within the sensitivity analysis found in section 3.1, five parameter couplings representing the most interesting outcomes were simulated for 50 years and stored as animations. The parameters chosen are:

- $NPP = 0.14, \epsilon = 0.35$
- $NPP = 0.18, \epsilon = 0.7$
- $NPP = 0.23, \epsilon = 0.07$
- $NPP = 0.28, \epsilon = 0.014$
- $NPP = 0.32, \epsilon = 0.007$

Animations can be found in the **GitHub** repository and are heavily recommended to get a clearer picture of the ecosystems dynamics. Snapshots are facilitated in figures 9, 10, 11, 12 and 13.

Time series graphs

The time series graphs extracted from the simulations made in the *sensitivity analysis* section can be found in figure 14.

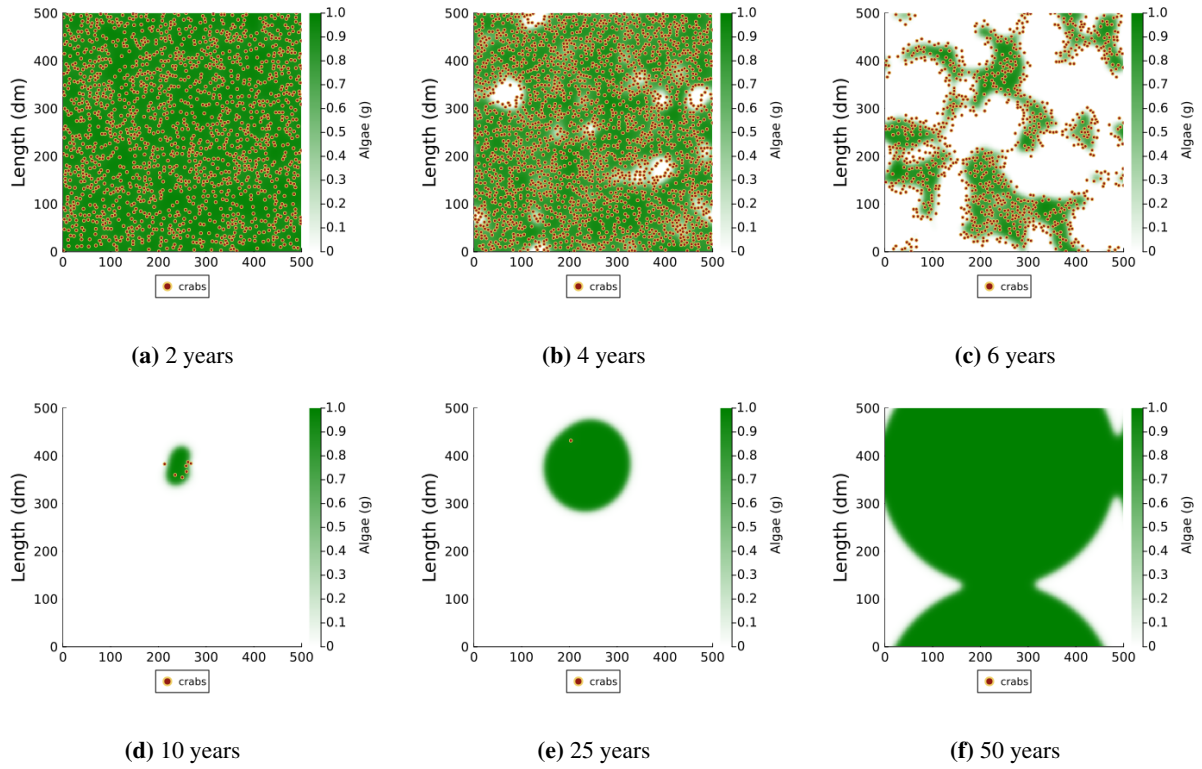


Figure 9: Snapshots of the herbivore-macroalgae model with parameters $NPP = 0.14$, $\epsilon = 0.35$.

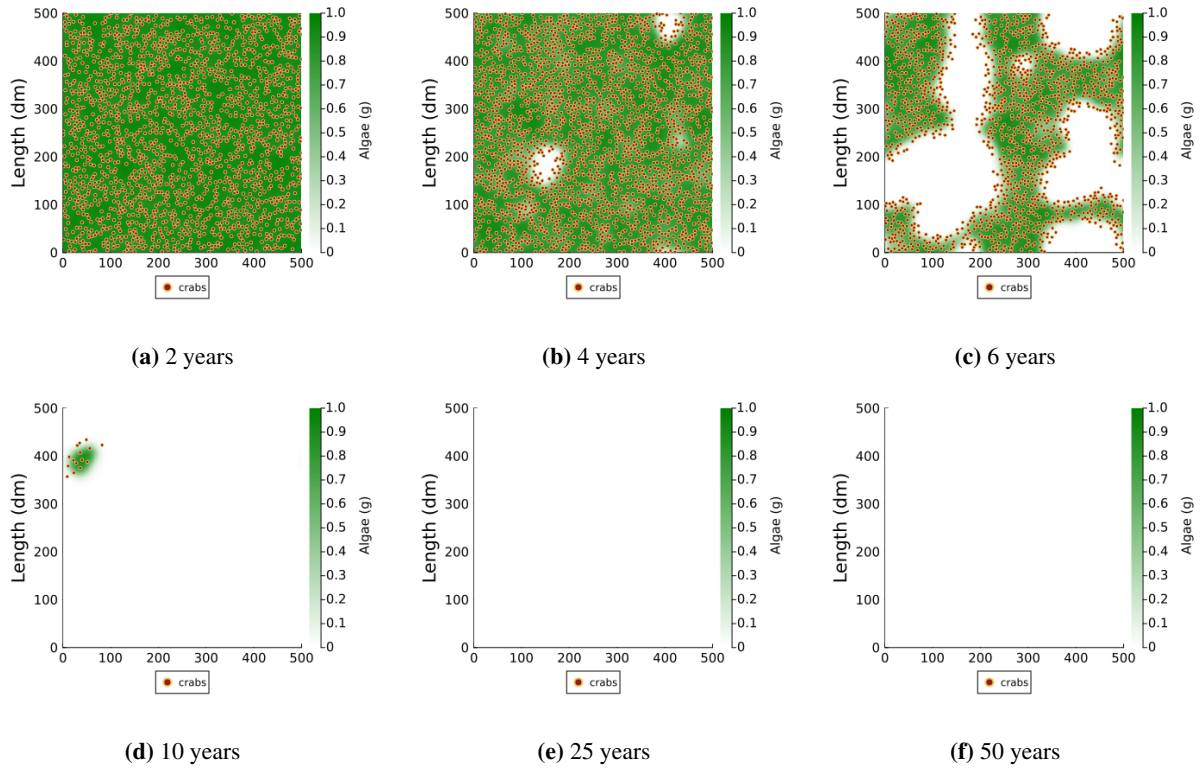


Figure 10: Snapshots of the herbivore-macroalgae model with parameters $NPP = 0.18$, $\epsilon = 0.7$.

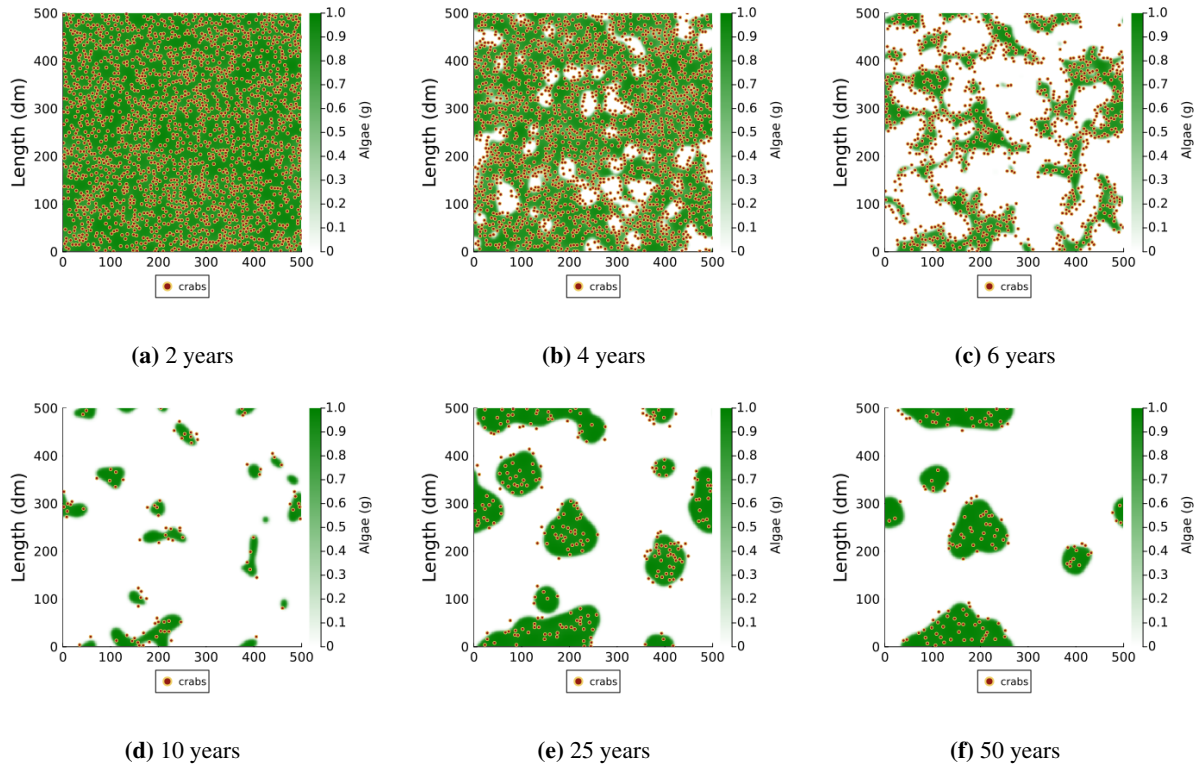


Figure 11: Snapshots of the herbivore-macroalgae model with parameters $NPP = 0.23$, $\epsilon = 0.07$.

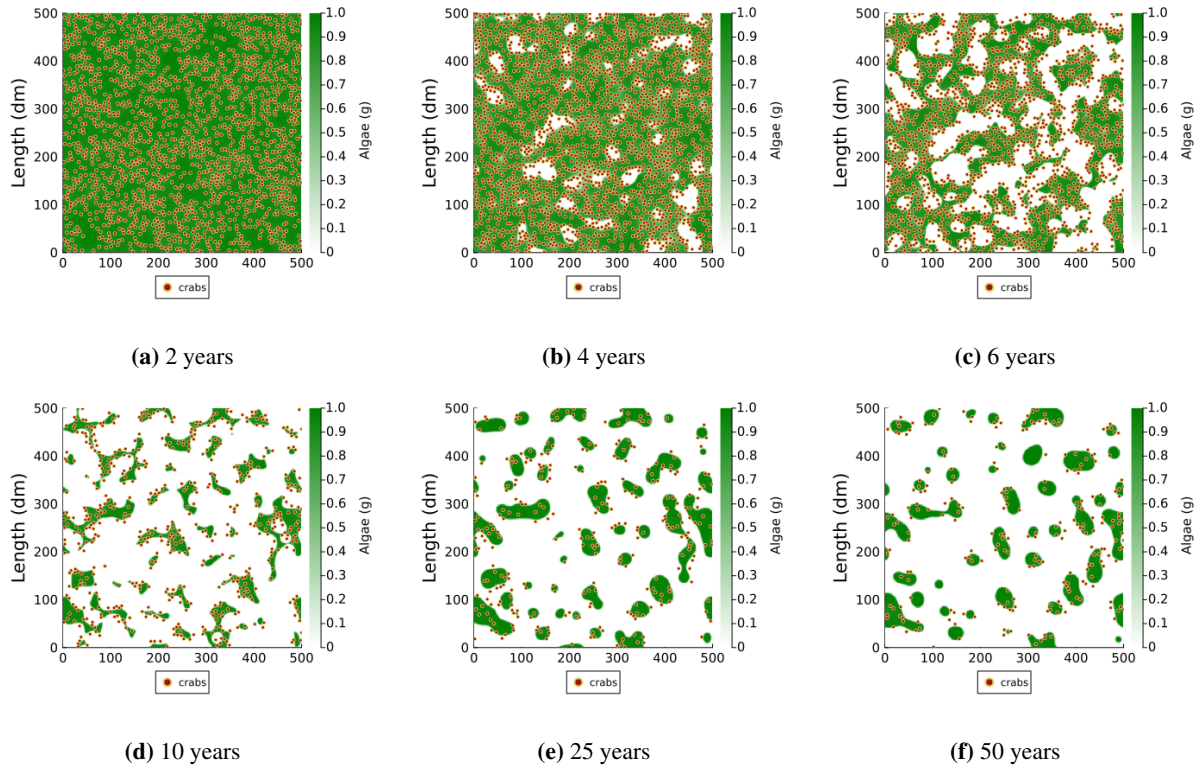


Figure 12: Snapshots of the herbivore-macroalgae model with parameters $NPP = 0.28$, $\epsilon = 0.014$.

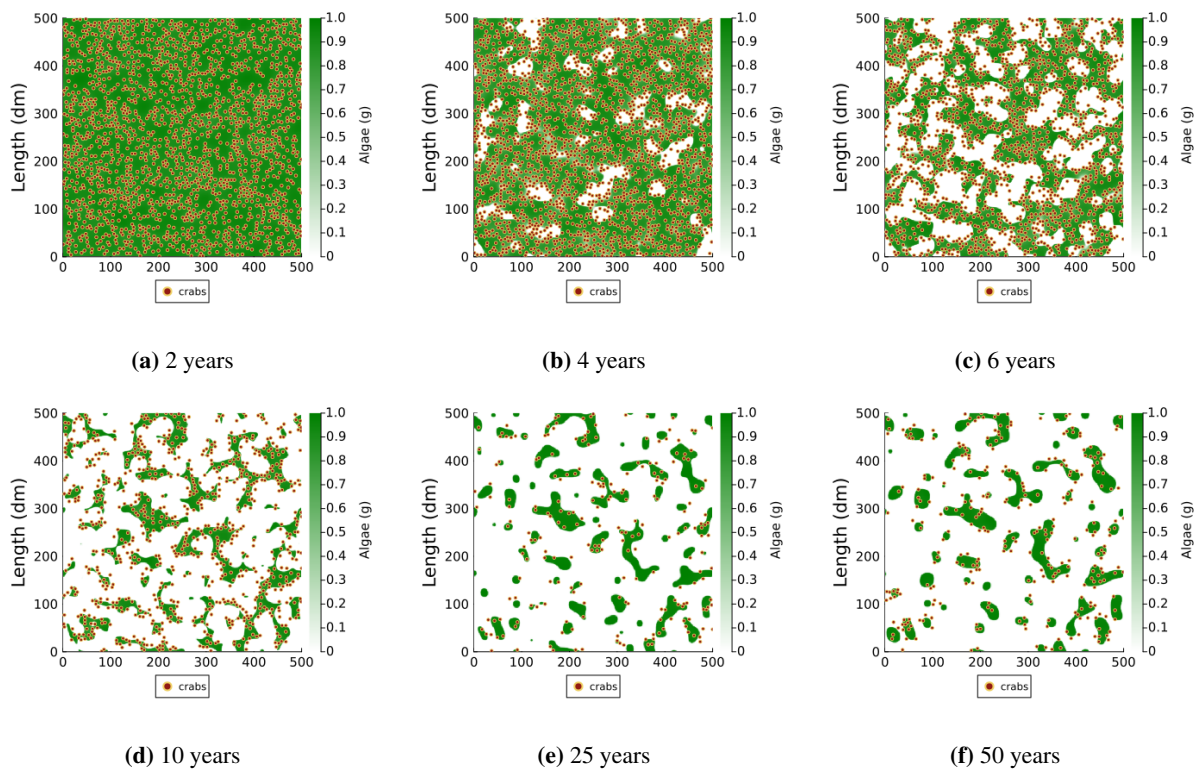


Figure 13: Snapshots of the herbivore-macroalgae model with parameters $NPP = 0.32$, $\epsilon = 0.007$.

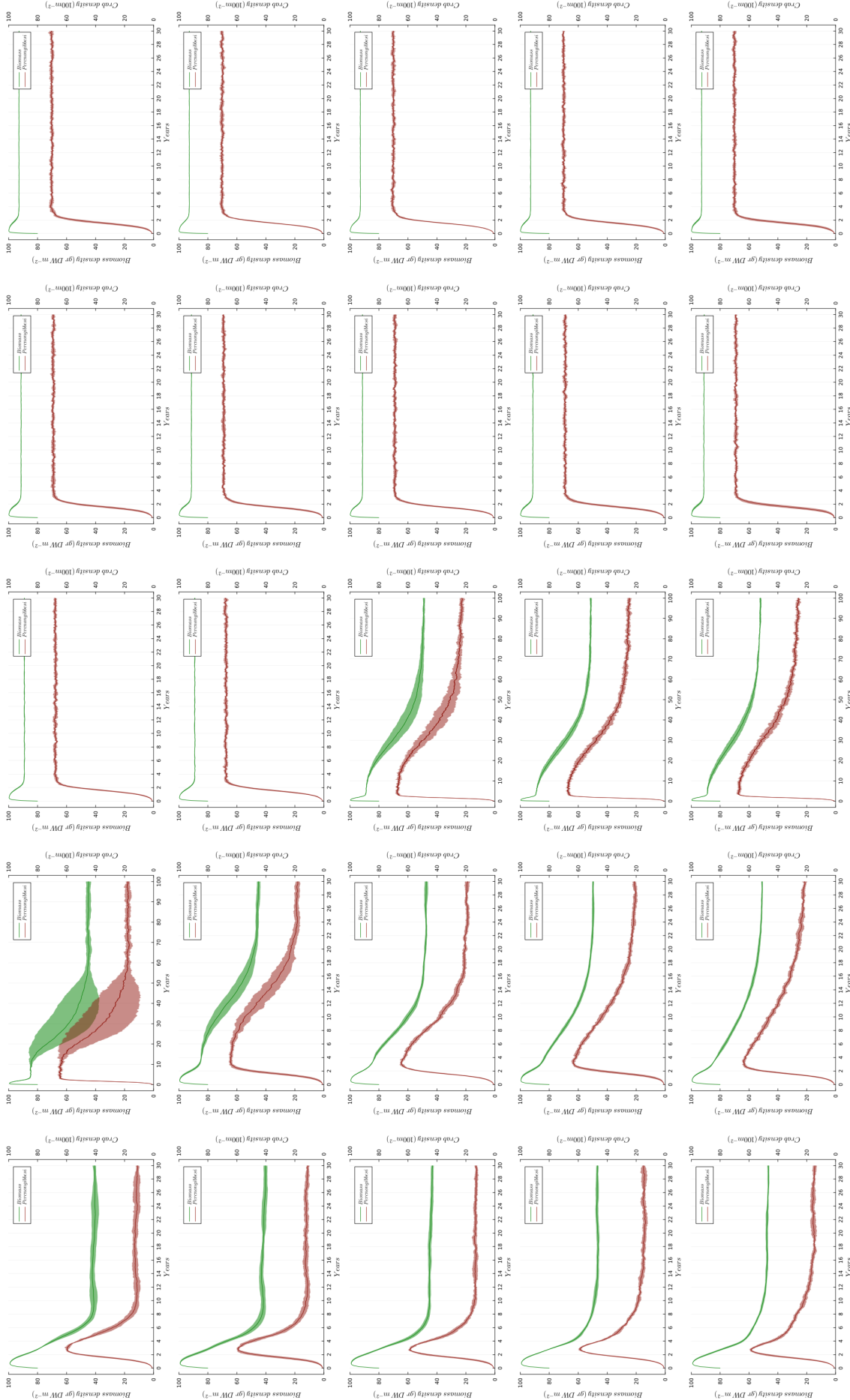


Figure 14: Compilation of graphs showcasing the time evolution of the ecosystem under different flora parameters. Algae density (g DW m^{-2}) is represented in green while crab density is represented in red (100 m^{-2}), both have a minimum of 0 and maximum of 100. The parameters mirrors the ones found in figure 4; NPP (columns, left-to-right: 0.14, 0.18, 0.23, 0.28, 0.32) and ϵ (rows, top-to-bottom: 0.7, 0.35, 0.07, 0.014, 0.007). Each combination of parameters is run 10 times to generate an average path and standard deviation represented as ribbons. Each run is simulated for 30 years with the exception of (b), (m), (r) and (w) which are simulated for 100 years.

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